# Twin-Pulse

**Cardiometabolic Digital Twin**

Between clinic visits, a

## 90-day gap

we can’t see what is happening

**Three critical information gaps**

What’s happening right now

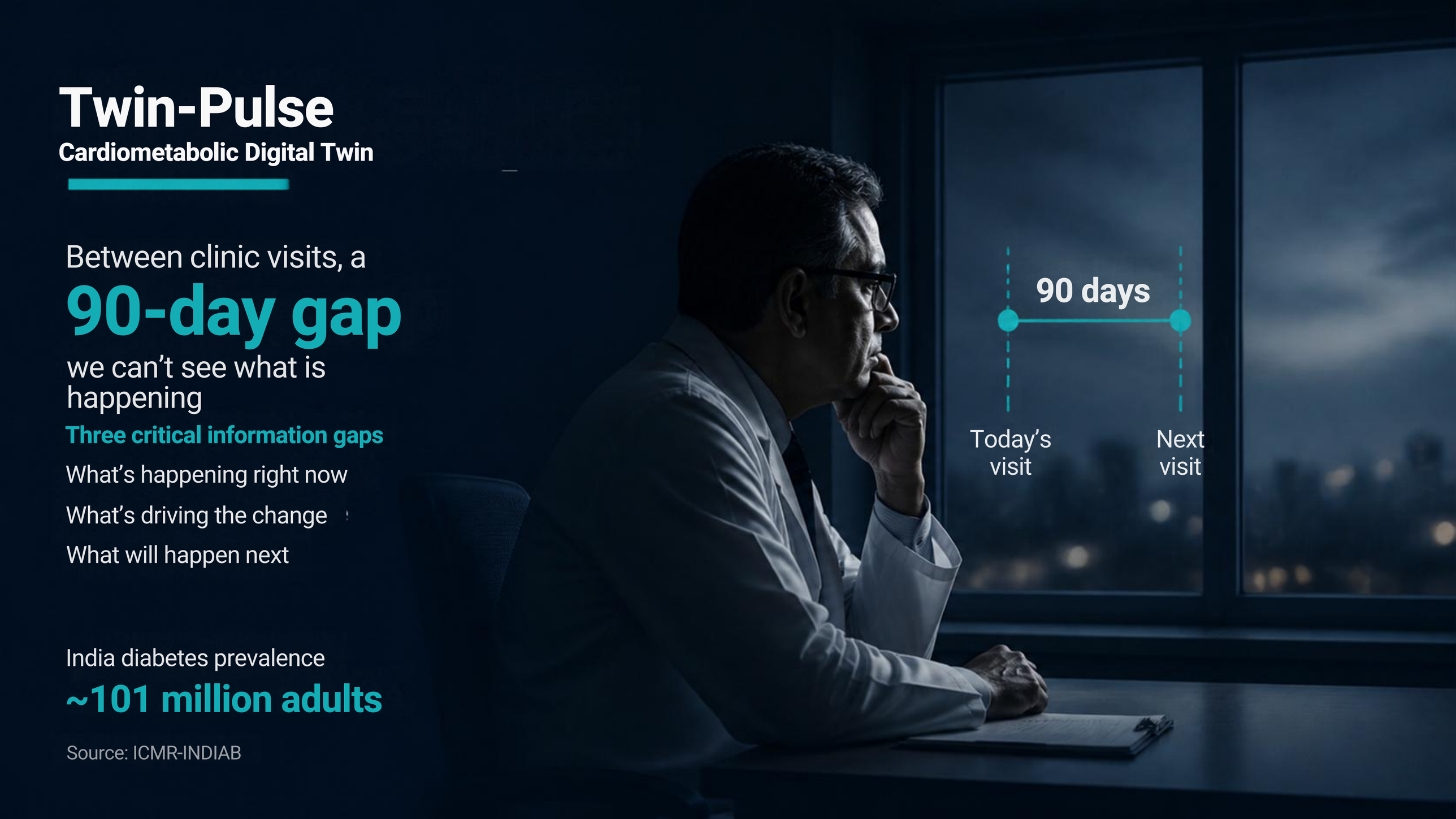
What’s driving the change

What will happen next

India diabetes prevalence

**~101 million adults**

Source: ICMR-INDIAB



# Objectives and need

## 01

### Predict a spike early

Detect deterioration sooner to give time for interventions that can change the outcome.

## 02

### Show the whole-body picture

Integrate vital systems and trends to reveal what's driving the risk.

## 03

### Let the doctor test actions safely

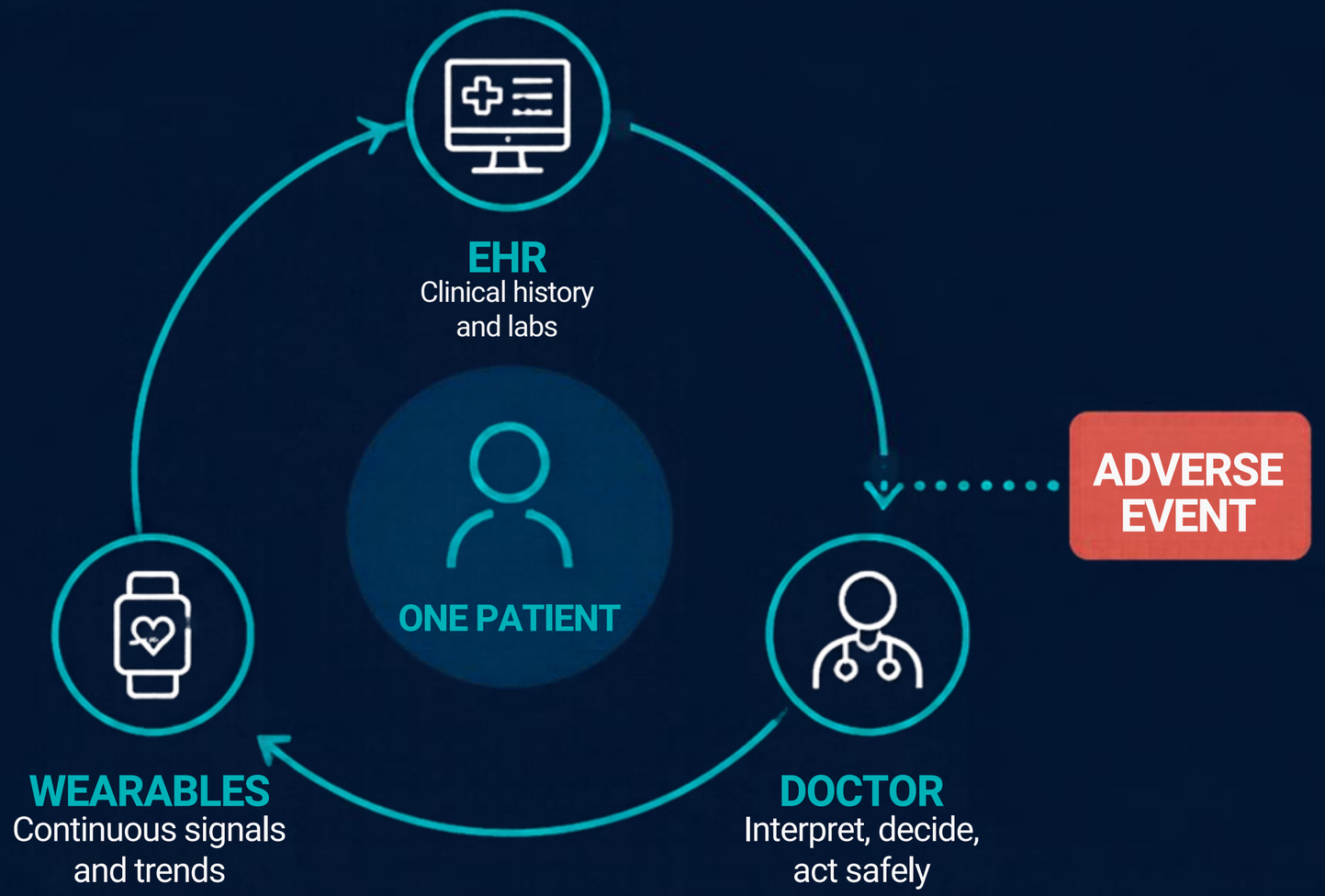
Explore possible actions and see likely impact before deciding at the bedside.

# Research and alternatives

| Twin-Pulse conceptually covers the full spectrum of what matters at the bedside. | Clinic HbA1c | CGM app | Risk score | Black-box ML | Twin-Pulse |
|---|---|---|---|---|---|
| Early deterioration signal | | ✓ | ✓ | ✓ | ✓ |
| Whole-body system view | | | | | ✓ |
| Explains why | | | | | ✓ |
| Tests actions safely | | | | | ✓ |
| Shows likely impact | | | | | ✓ |

All approaches have value. This is a conceptual comparison; no direct head-to-head benchmark exists.

# One patient. Three questions.



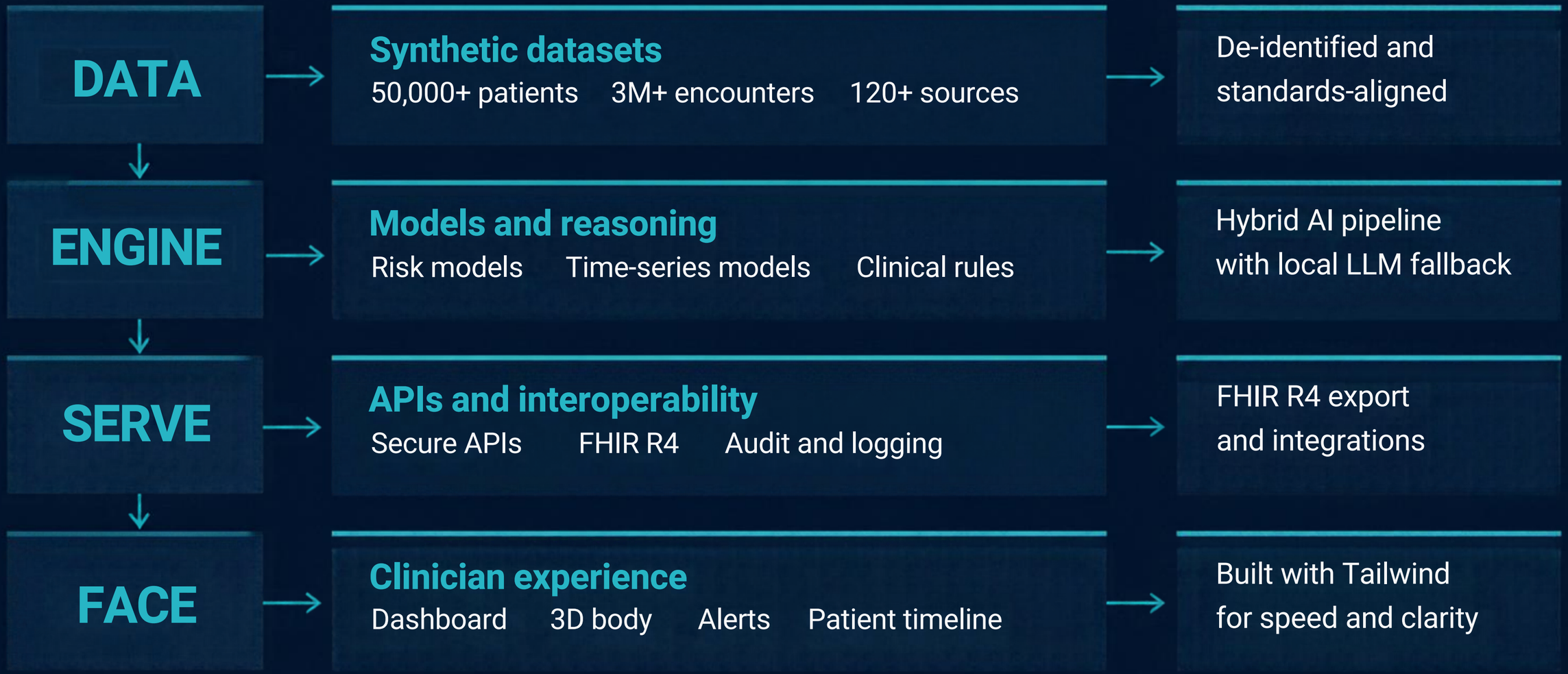

01 **NORMAL**
How long until they cross the line?

02 **EARLY DETERIORATION**
Will it keep getting worse?

03 **CRITICAL THRESHOLD**
Will it become life-threatening?

04 **DECOMPENSATION**
What happens next, and how do we act safest?

# Architecture and stack



# Outputs and synthetic test results

SYNTHETIC EXAMPLE

## Rajesh Sharma

| | |
|---|---|
| Spike risk (15 min) | 0.86 |
| Time to peak | 32 min |
| Predicted peak | 212 mg/dL |
| Suggested action | +2U insulin |
| Confidence | High |

SYNTHETIC CONFUSION MATRIX

| Actual \ Predicted | Spike | No spike |
|---|---|---|
| Spike | 198 | 22 |
| No spike | 28 | 52 |

| | |
|---|---|
| Sensitivity | 88% |
| Specificity | 70% |
| Precision | 81% |
| F1 score | 0.84 |

**+2U insulin is a synthetic fixture for testing only. This is not treatment advice.**

300 synthetic test windows

# Impact, Limits and next steps

## FOUR STAKEHOLDER BENEFITS

### Better decisions

See risk earlier and act with greater confidence.

### Safer care

Prevent deterioration and reduce unnecessaryEscalations.

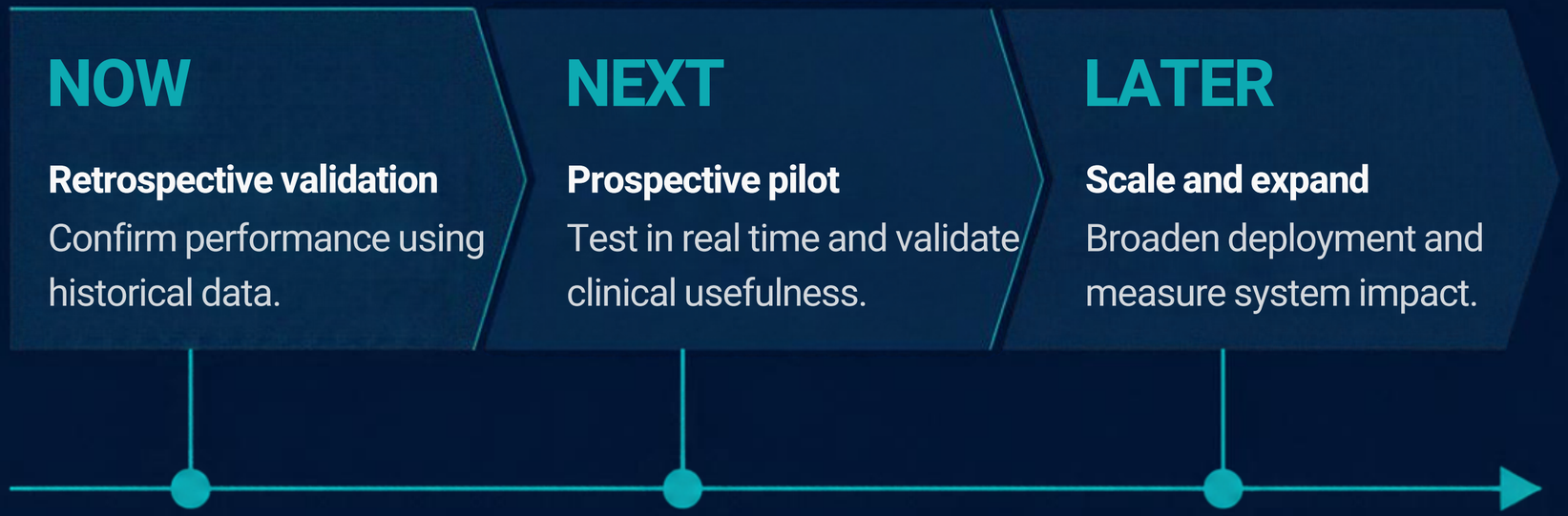
### More efficient systems

Focus time and resources where impact is greatest.

### Clearer evidence

Turn data into actionable insight that drives outcomes.

## VALIDATION ROADMAP




From **reactive visits** to **continuous foresight**.